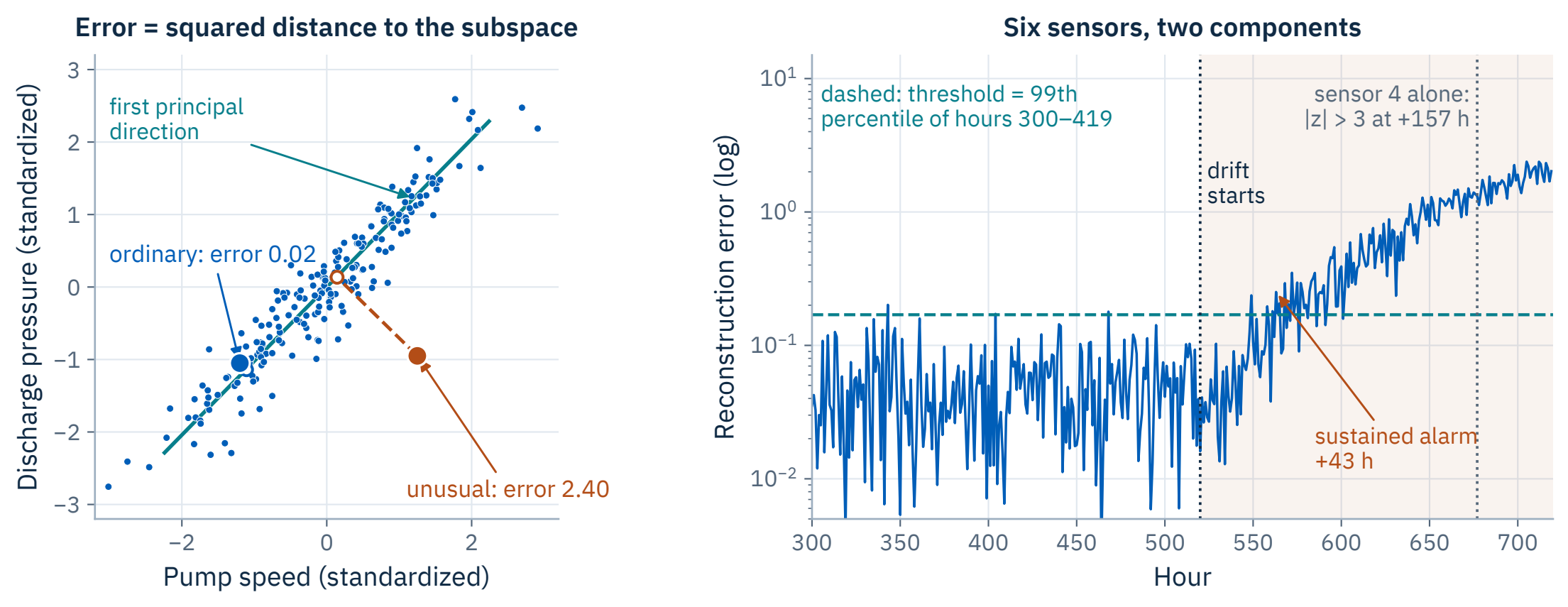
Error = squared distance to the subspace
first principal direction
ordinary: error 0.02
unusual: error 2.40
Discharge pressure (standardized)
Pump speed (standardized)
3
2
1
0
−1
−2
−3
−2
0
2
Six sensors, two components
dashed: threshold = 99th percentile of hours 300–419
sensor 4 alone: |z| > 3 at +157 h
drift starts
sustained alarm +43 h
Reconstruction error (log)
10^{1}
10^{0}
10^{-1}
10^{-2}
300
350
400
450
500
550
600
650
700
Hour